

Design and Test of the Smart Composter Controlled by Sensors

Mustafa Elalami*, Yassine
Baskoun, Fatila Zahra
Beraich, Moha Arouch
Laboratory IMII, Faculty of
Science and technical
Hassan 1 University
Settat, Morocco
mustafa.elalami@gmail.com

Mohamed Taouzari
Laboratory LISA, high School of
technology Berrechid
Hassan 1 University
Settat, Morocco

Salah Dine Qanadli
Cardio-Thoracic and Vascular Unit,
Department of Radiology
University Hospital of Lausanne
Lausanne, Switzerland

Abstract— Treatment of organic wastes in Morocco remains a sector that consumes large budgets and which increases with demographic growth without finding a targeted and relevant solution until today. In this work we present the study, the design and the realization of a new composter destined to convert the residues food and all type of organic wastes in situ (to eliminate transport charges) to obtain a mature compost with high quality in 4 weeks. This system is based on the intern design and automated and optimized control of all operating parameters. This solution will reduce up to 80% of waste produced per day in the city and will be an effective tool for the provincial managers; the prototype has been validated in the field.

Keywords— design; realization; composting machine ; remote control; smart composter; sensors

I. INTRODUCTION

For several years the landfill has been the most prevalent method of disposing of household and similar waste in Morocco as all developing countries because of its simplicity, but also because of its lower cost than other methods. There is a huge amount of waste produced every day and which contains more than 75% of the organic fraction. If this fraction does not have be treated in time it begins to decompose naturally in the absence of oxygen in the dumps the organic matter starts to generate biogas composed mainly of carbon dioxide and methane that are the main cause of global warming. Add to this the flow of leachate that contaminates the water table.

One of the perfect waste treatment techniques is composting, that's the first way employed by our ancestors, because of its related benefits also it's the natural way [1] to recycling organic residues such as hygienization, profitability and the transformation of waste into value-added products "Ref. [2]".

Composting is a reliable treatment of organic waste that could be useful in reducing the negative effects that can result from the spreading of organic waste on land [3] as it provides disinfected and stabilized products that could be employed as a potential source of organic fertilizer or in soil amendments [1]. The composting process is influenced by physicochemical factors like the temperature, pH, particle size, moisture content, aeration [4]. Composting is a biochemical and heterogeneous process implying the mineralization of organic matter into CO₂, NH₃, H₂O and

incomplete hydration to produce a stabilized final product with reduced toxicity and pathogenic organisms [3].

Considering the benefits of composting, it can be employed to transform organic waste into useful products by using the degradation attributes of the macros and microorganisms present in the contaminant matrix. It is a technology widely used to transform organic waste into organic fertilizer (compost), recycling nutrient minerals (nitrogen (N), phosphorus (P) and potassium (K)) that can be employed for agricultural purposes [5].

However, the composting process can only be performed if it is well managed [6] and requires precise monitoring and control to obtain useful information in real time [7]. Therefore, a successful composting operation requires good mixing ratios [8] as well as the optimization of different process parameters such as the aeration rate and an appropriate ratio (carbon / nitrogen content and content) [9]. The process of accelerating composting requires the use of a rotating system with active aeration to accelerate the degradation of waste and reduce the active composting phase "Ref. [10, 11]" as well as to control and manage all composting parameters in real time using a remote control.

Objective of this work design and the manufacturing of an electromechanical machine dedicated to treatment of organic waste and managed by a remote control that allows all composting parameters to be controlled in real time.

II. MATERIAL AND METHOD

A. Description of the machine composting :

The urban composting prototype below "Fig. 1" is designed as a closed container capacity 0.930 m³. The rotary drum composting system (2) destiny for processing food residues, agricultural residues and any organic effluent by an aerobic fermentation process completely mixed by the rotation of the drum and the steel sheets (18) welded with inside the drum that is equipped by an active aspiration (4, 5, 21) "Fig. 1(E)". The rotating drum is constructed from stainless steel sheet and supported by a frame (1) (black steel rectangular tube 60x40 mm), two doors (10) "Fig. 1 (E)" located the on part longitudinal the drum (500, 450 mm) are used for the loading's / unloading's of solid organic waste, a rubber gasket lining is fixed on the inner side of each door keep it watertight and prevent leakage. Aeration of drum is achieved by using a suction centrifugal fan with variable flow (4) and perforated holes (21) located at the lateral part

of the right side of the drum “Fig. 1 (E)” to homogeneous introduction of air into the drum. Air outlet pipe (5) is located at the left side of the drum.

The drum is guided in rotation by two bearings (6,7) “Fig. 1, (D), (C)”, and rotates at a speed of (4tr / min) by an electric motor and a reducer with gear train with helicoidally teeth (3) “fig.1, (A)” and a belt pulley system (8, 9) “fig.1, (A), (D)”. Temperature, humidity, NH₃ content, PH level and oxygen level are checked by means of sensors installed regularly on the round tube (13) “fig.1, (A)” that is fixed on the tube (5) and supported by a plain bearing “fig.1, (A)”. Data are recording using a PLC with remote management module installed in a control cabinet (15) “Fig.1, (B)”.

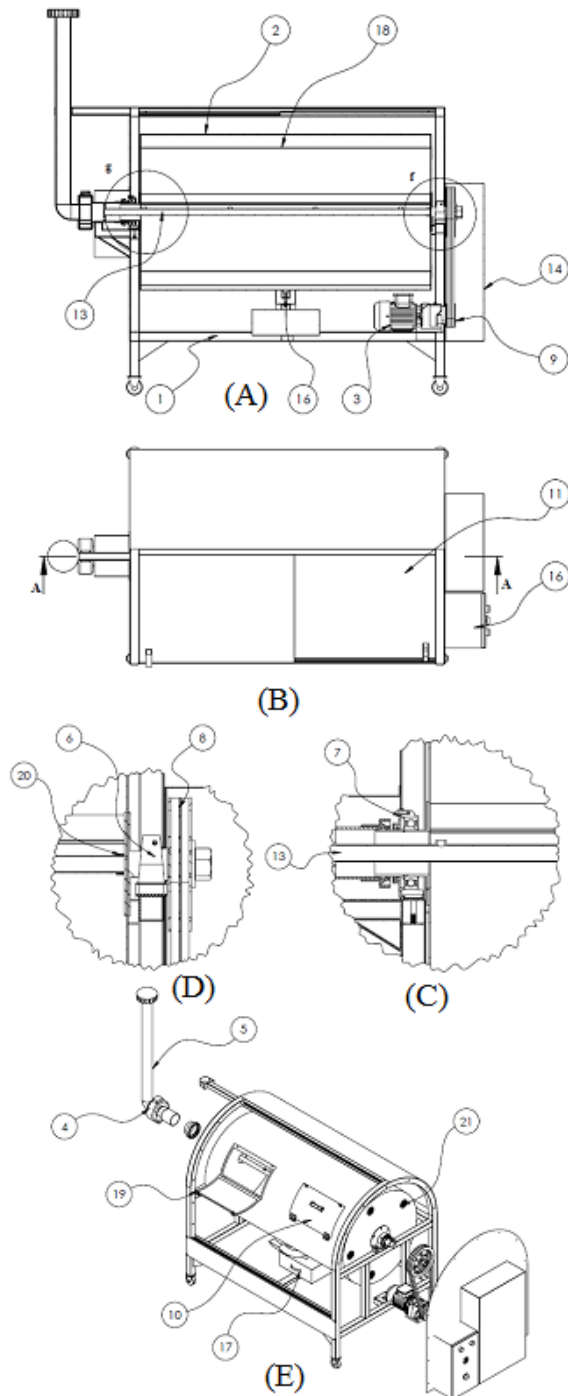


Figure 1. Assembly drawing of the rotating drum composting machine

Where : (A) Cross-section front view A-A ; (B) Top view of the machine; (C) Detail view g for connection between the left side of the drum and the frame; (D) Detail view f for connection between the right side of the drum and the frame; (E) Explode view drawing of machine composting.

B. Description of the remote controle :

The system will collect data from the rotary composter such as (temperature, humidity, NH₃ ammonia and PH) and download it to the bridge. After receiving the data, the server-based cloud platform will store the data in the database “Fig. 2” .Then the client (including the machine control terminal, mobile phone application and the browser of the remote computer) can acquire data from the database in a timely manner, which can appear as a number or a curve.

The user of the application can send instructions to the composting machine in case he finds that the parameters detected by the application are not suitable.

The gateway will integrate the data and use the 4G network to transfer the data to the Internet server cloud platform of the assigned IP address.

For example, if the humidity exceeds 60%, machine rotated automatically and starts aeration (instructions of auto-control are shown as in TABLE I). In addition, the users can use the software to control devices directly on the control terminal of the composting machine.

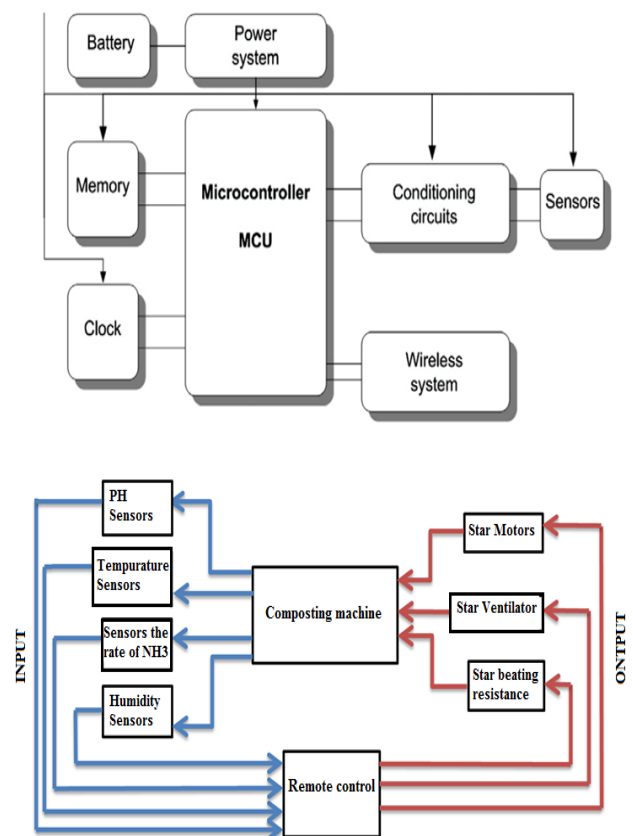


Figure 2. Block diagram of remote control

TABLE I. CONTROLLER AUTO CONTROL INSTRUCTIONS

Composting parameter	causes of the problem	Auto-control setup
NH ₃ >20ppm	Mixture too rich in nitrogenous matter Excessive temperature (over 70 °)	Activate rotation Activate aeration Stop heating resistance
Humidity <40%	Lack of water	Active manual command (inject the water)
Humidity >60%	Lack of nitrogenous	active rotation Activate aeration
Temperature <20°	Lack of nitrogenous	Activate heating resistance Activate rotation Activate aeration
Temperature >65°	Heating resistance activate	Activate rotation Activate aeration Stop heating resistance

C. Dimensioning of the rotary composting machine:

1) Characteristic of the rotating drum:

The rotary drum is a hollow cylinder with two flat bottoms of dimensions (length L, diameter D, thickness e). It rotates at a rotational speed (1 tr/min to 4 tr/min), supported by two bearings, and it is made of stainless steel whose yield strength (Re = 172.34 MPa).

2) Characteristics of organic waste:

Organic waste is food waste, green waste, agricultural waste, and waste from the food industry or the catering industry. It is characterized by an average humidity rate of 50% and density ρd see TABLE II.

3) Determination of the thickness of the drum e:

To avoid warping and drum failure the choice of a minimum thickness of the drum is necessary. This drum thickness is calculated according to the load, the geometry and the material by the equation:

$$\sigma_{\theta} = \frac{D \cdot P_i}{2e} \quad (1)$$

TABLE II. THE DENSITY OF ORGANIC WASTE

Types of material	Minimum densities in (kg/m ³)	Maximum densities in (kg/m ³)
Branches	120	180
Crushed branches	230	330
Fresh lawn	350	500
Fresh mixed waste (branches and grass) not ground	250	300
Fresh mixed waste (branches and grass) ground	350	450
Old grass in a bucket	550	700
Dead leaves	300	400
Fresh green grass	350	450
Hay (dry grass)	120	180
Dry grass	180	250
Dry restaurant waste	500	600
Wet restaurant waste (washing)	850	1000
Horse manure (fresh)	150	220
Horse manure (old)	400	500

Condition of resistance:

The circumferential stress must be less than or equal to the elastic limit, the relation is given by the inequation:

$$\sigma_{\theta} = \frac{D \cdot P_i}{2e} \leq \frac{Re}{s} \quad (2)$$

The minimum thickness is calculated by the relation “(1)”, “(2)” by the following equation:

$$e = \frac{D \cdot P_i \cdot s}{2 Re} \quad (3)$$

Where:

σ_{θ} : Circumferential stress in (Mpa)

D : Diameter of the rotating drum in (mm)

P_i : Of the mass of the load on the occupied surface in

(N / mm²)

s : The safety factor equals 1.5 for steel structures

Re: elastic limit in (Mpa)

4) dimensioning of the rotary drum drive system:

The drum makes a rotational movement based on its axis of rotation subjected to a load of organic waste at a density of ρ in (kg/m³). The drum is filled with (2/3) of its volume to facilitate the degradation of organic waste.

a) Determination of the torque required to drive the drum:

Torque is calculated as a function of the mass of organic waste and total mass of the drum and the distance from the axis of rotation by the following equation:

$$C_t = \left[\frac{\pi}{12} \cdot \rho d \cdot g (L - 2e) (D - e)^2 + 4M_t \right] \cdot (L - 2e) \quad (4)$$

Where:

C_t : Torque to drive the drum in (N.m).

ρd : Density of organic waste in (kg/m³).

L: Length of the drum in (mm).

M_t : Total mass of the rotary drum in (Kg)

According to equation “(4)”, the mechanical rotational power to drive the drum is determined by the following equation:

$$P_m = \frac{1}{9550} N_t \cdot C_t \quad (5)$$

Where:

P_m : Mechanical power of rotation in (KW)

N_t : Rotational speed in (tr/min)

b) Determination of the electrical power for the choice of the electric motor:

Rotational speed of the drum is very slow (1 tr/min to 4 tr/min) compared to the speed of the Electric Motor. This last were chosen with a rotational speed (1500 tr/min) and to reduce speed a gearbox are chosen by (ratio reduction r_1 , efficiency η_1) and a belt pulley system (ratio reduction r_2 , efficiency η_2). Useful power of the electric motor P_u is obtained by function “(4)”, “(5)” and the total efficiency of the transmission system is determined by the following equation:

$$P_u = \frac{P_m}{\eta_1 \eta_2} \quad (6)$$

c) *Dimensioning of the V-belt pulley:*

The V-belt pulleys are dimensioned according to standards using the machine element dimensioning book [12]. The axes of two pulleys are parallel, space limitations require that the center distance C is approximately (600mm) and the pulleys are optimal. The driving diameter D_m is set at a minimum value (65 mm) and the primitive diameter for the receiving pulley D_r must be less than C

$$(D_r < C < 3(D_m + D_r)).$$

$$\text{Reduce Ratio } r_2 : r_2 = \frac{D_m}{D_r} \quad (7)$$

To obtain dimensions of the pulley and the belt we must determine the winding angle of the belt θ and the length of the belt L_c , the gross power P_r , the net power transmitted by the belt P'_r , the effective power P' and the number of belts n_c :

$$\theta = \pi - \sin^{-1} \frac{(D_m - D_r)}{2C} (\text{rad}) \quad (8)$$

$$L_c = 2C + \frac{\pi}{2}(D_m + D_r) + \frac{(D_m - D_r)^2}{4C} \quad (9)$$

$$P_r = \left[c_1 - \frac{c_2}{D_m} - c_3(r.D_m)^2 - c_4 \log(r.D_m) \right] r.D_m \quad (10)$$

$$+ c_2 r \left[1 - \frac{1}{Ka} \right] \quad (10)$$

$$P'_r = K_1 K_2 P_r \quad (11)$$

$$P' = K_s \frac{P_u}{\eta_1} \quad (12)$$

$$n_c = \frac{P'}{P'_r} \quad (13)$$

Or c_1 , c_2 , c_3 and c_4 are constants that are a function of the belt section and Ka is a constant depends on speed ratio “(4)”, and r represents the rotational speed of small pulley divided by 1000, K_1 is a factor applied to the belt winding and 2 is a factor applied to the length, K_s is an overload factor in use.

d) *Reducer dimensioning :*

Reducer with gear train with helicoidally teeth unit reduces the rotational speed of the motor and increases the force. The reduction ratio must be calculated according to the reduction ratio of the belt pulleys and the overall ratio of the drive system.

$$r_1 = \frac{N_t}{r_2 N_m} \quad (14)$$

D. *Determination of the energy consumption of the machine:*

The composting machine is characterized by the dimensions of the rotary drum (length 1500 mm, diameter 900 mm, thickness 3mm) and its mass (140 Kg). The mass of organic waste is (217 Kg) and rotation speed (4tr / min).

Daily energy consumption is calculated according to the energy absorbed by the machine by equation “(6)”, where the machine working 3 hours a day. The value of the absorbed power equals 1kW considering two transmission system efficiency ($\eta_1 = 0.95$, $\eta_2 = 0.95$).

The monthly energy consumption by the machine:

$$EC = 30.3.P_u = 90KWh$$

III. RESULTST AND DISCUSION

The design and mechanical study was carried out by the laboratory team of Engineering, Industrial Management and Innovation “LIMII”. The realization is carried out in collaboration with the company Biodome and we are based in this study on the telemanagement of the parameters of composting and we receive all the information from the composter to a smartphone and a computer via 4G and Wifi. Design and realization of the machine shown in “fig. 3”.

We receive information about moisturize, temperature, airflow, drum rotation or stoppage, and we act by telemanagement of on/off of the system, drum rotation, rotation time and waiting time, water injectors in case of dry compost, air blowing and suction in case of excess moisture, action on heating activation. The following table illustrates the parameters and the action to be taken for each situation “Fig. 4”. Manual actions are limited in the feeding of the composter by carbon and nitrogenous waste. The telemanagement system of the machine allows managing all the composting parameters in real time via an application installed on a smartphone “fig. 5”.



Figure. 3. Real composting machine



Figure. 4. Smart composting machine

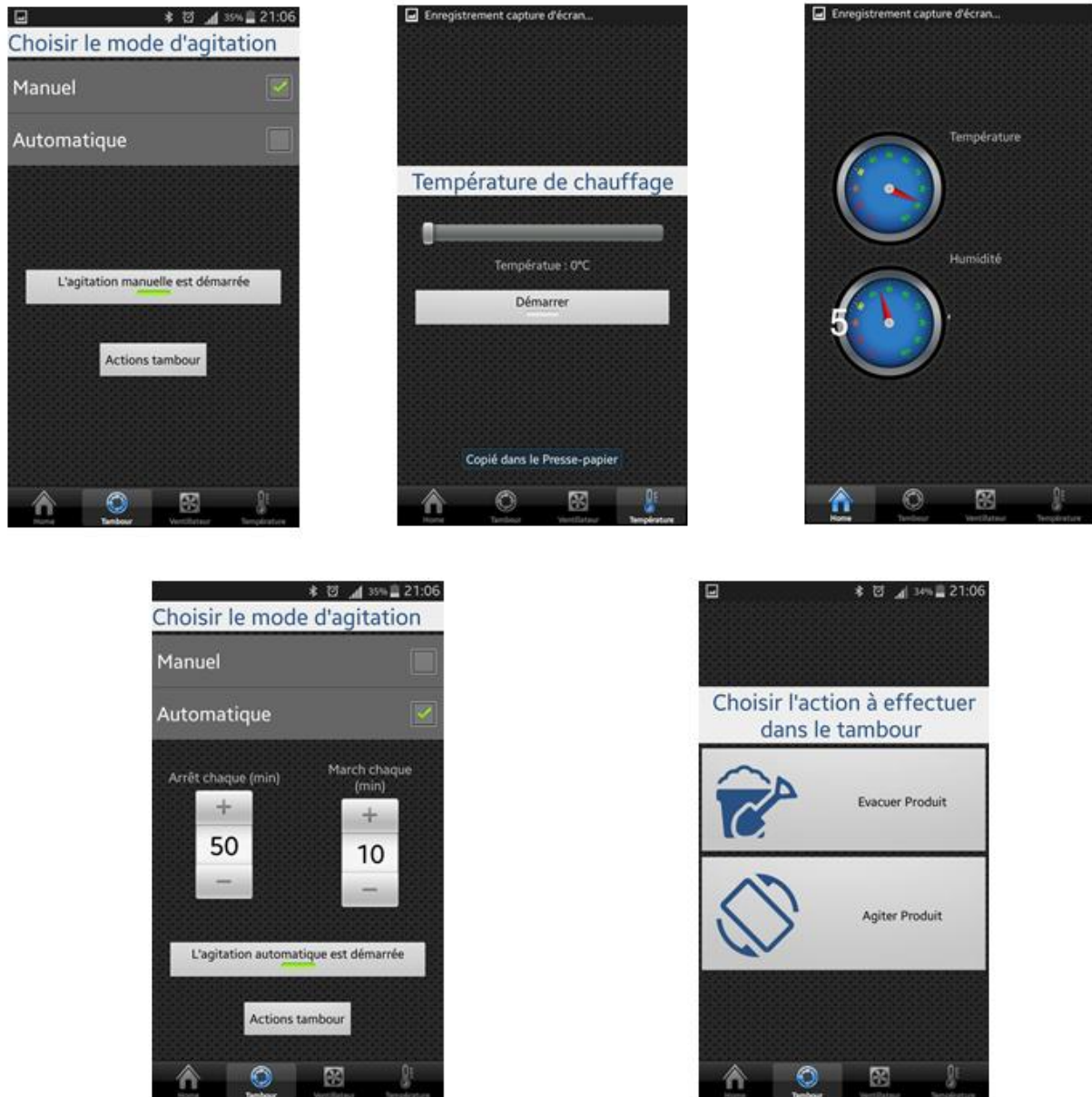


Figure 5. Application interface in smartphone

IV. CONCLUSION

This document reports study, design and implementation of a composting machine and a remote control designed to provide data to facilitate control of the composting process and ensure the main composting parameters. The use of telemanagement in composting reduces the composting time in 4 weeks thus produce mature compost.

The remote management integration in the composting machine requires extremely robust equipment both mechanical and electrical, who presented the following design and implementation challenges from the mechanical and electrical / electronic point of view. : (a) installation of sensors in a rotary drum; (b) measurement of temperature, humidity, NH₃ rate, PH rate with sensors; (c) wireless communications from the user application / to the machine.

The solutions described in the previous sections have worked adequately, meeting the design specifications. Prototype of the system, which is an advanced innovative solution for controlling and managing composting, has been tested in a very positive way. Although there are improvements to be made such as the addition of photovoltaic panels for power supply, we have reached a phase of marketing

ACKNOWLEDGMENT

Research leading to the above results received funding from the company BIODOME.

REFERENCES

- [1] X. Qian, G. Shen , Z. Wang, C.Guo, Y.Liu , Lei.Z, Z Zhang, "Co-composting of livestock manure with rice straw, Characterization and establishment of maturity evaluation system," *Waste Manage*, vol. 34, pp. 530-535, February 2014.
- [2] I. Heribert G.B. Maria , A.J. Ascher , " Recycling of organic Wastes to Soil and its Effet on Soil Organic Carbon Status," *The Feature of Soil Carbon*, pp. 195-214, April 2018.
- [3] M. Das, H.S.Uppal , R.Singh, S.Beri, K.S. Mohan, V.C.Gupta, A. Adholeya , "Co-composting of physic nut (*Jatropha curcas*) deoiled cake with rice straw and different animal dung, " *Bioresour. Technol*, Vol. 102 (11),pp. 6541-6546, june 2011.
- [4] M.F.D. Juarez, B. ahauser, A. Walter, A., Insam, I.H. Franke-Whittle, "Cocomposting of biowaste and wood ash, influence on a microbially driven-process," *Waste Manage*, vol 46, pp. 155-164, december 2015.
- [5] Y.Wang, , P.Ai, H. Cao, Z. Liu, , "Transformation of organic matters in animal wastes during," *Bioresour. Technol*, vol 193, pp. 200-205, decembre 2015.
- [6] M.K. Awasthi, A.K. Pandey, J. Khan, P.S. Bundela, J.W.C. Wong, A. Selvam, "Cocomposting of gelatin industry sludge combined with organic fraction of municipal solid waste and poultry waste employing zeolite mixed with enriched nitrifying bacterial consortium. *Bioresour. Technol.*," *Bioresour. Technol*, vol. 213, pp. 181-189, august 2016.
- [7] F. Storino, J.S. Arizmendiarieta, I. Irigoyen , J.Muro , P.M. Aparicio-Tejo, "Meat waste as feedstock for home composting: effects on the process and quality of compost, " *Waste Manage* , vol. 56, pp. 53-62, october 2016.
- [8] K. Kazemi, B. Zhang, L.M. Lye, Q. Cai, T. Cao, " Design of experiment (DOE) based screening of factors affecting municipal solid waste (MSW) composting," *Waste mange*, vol. 58, pp. 107-117, december 2016.
- [9] B. Zang, S. Li, F. Jr. Michel, G. Li, Y. Luo, D. Zhang , Y. Li, "Effects of mix ratio, moisture content and aeration rate on sulfur odor emissions during pig manure composting, " *Waste Manage* , vol. 56, pp. 498-505, december 2016.
- [10] F.N. Alkoaik , " Integrating aeration and rotation processes to accelerate composting of agricultural residues, " *PLOS ONE* 14(7): e0220343, july 2019.
- [11] F.Z. Beraich , M. Arouch , M. Bakasse. Patent MA 37240 A1, july 2016.
- [12] G. Drouin, M.Gou, P.Thiry, R. Vinet , machine elements. Montréal : Polytechnic School of Montreal, 1986.